Milk Consumption During Teenage Years and Risk of Hip Fractures in Older Adults

Importance Milk consumption during adolescence is recommended to promote peak bone mass and thereby reduce fracture risk in later life. However, its role in hip fracture prevention is not established and high consumption may adversely influence risk by increasing height. Objective To determine whether milk consumption during teenage years influences risk of hip fracture in older adults and to investigate the role of attained height in this association. Design Prospective cohort study over 22 years of follow-up Setting United States Participants Over 96,000 Caucasian postmenopausal women from the Nurses' Health Study and men age 50 and older from the Health Professionals Follow-up Study Exposures Frequency of consumption of milk and other foods during ages 13–18 and attained height were reported at baseline. Current diet, weight, smoking, physical activity, medication use, and other risk factors for hip fractures were reported on biennial questionnaires. Main Outcome Measures Cox proportional hazards models were used to calculate relative risks (RR) of first incident hip fracture from low-trauma events per glass (8 fl oz or 240 mL) of milk consumed per day during teenage years. Results Over follow-up, 1226 hip fractures were identified in women and 490 in men. After controlling for known risk factors and current milk consumption, each additional glass of milk per day during teenage years was associated with a significant 9% higher risk of hip fracture in men (RR=1.09, 95% CI 1.01–1.17). The association was attenuated when height was added to the model (RR=1.06, 95% CI 0.98–1.14). Teenage milk consumption was not associated with hip fractures in women (RR=1.00, 95% CI 0.95–1.05 per glass per day). Conclusion and Relevance Greater milk consumption during teenage years was not associated with a lower risk of hip fracture in older adults. The positive association observed in men was partially mediated through attained height.